Adı Soyadı:
Numarası:
İmza:

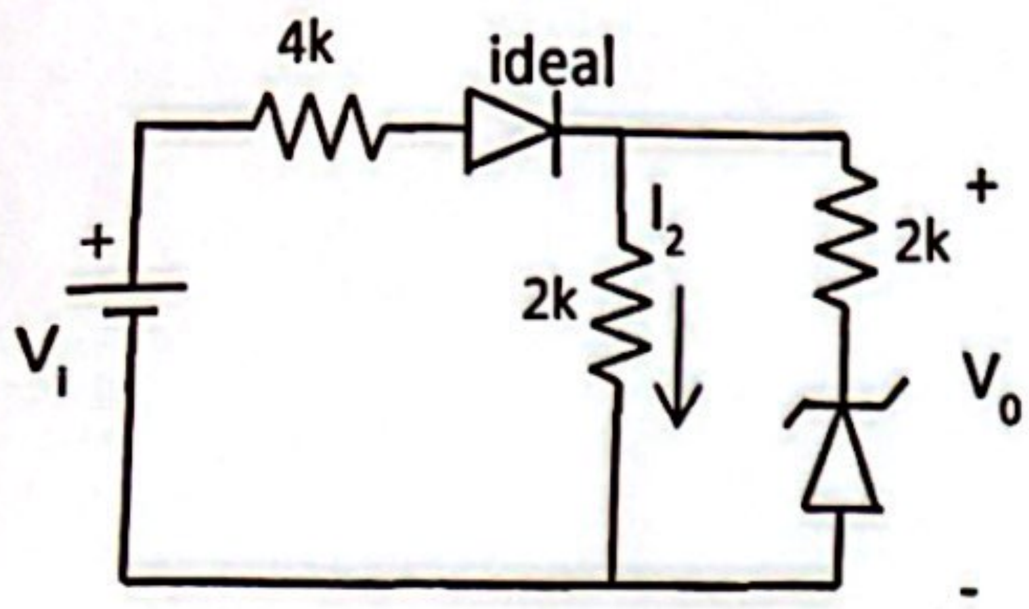
T.C. Konya Teknik Üniversitesi, Müh. ve Doğ. Bil. Fak., Elk.-Elt. Müh. Böl. Elektronik I Dersi VİZE Sınavı

1	2	3	4

Tarih:29.11.2023 **Süre:**90 dak.**BAŞARILAR…** Prof.Dr. Seral ÖZŞEN & Prof.Dr. Murat CEYLAN

Ad, soyad, numara bilgilerini eksik ya da hatalı giren öğrencilerden 5 puan kırılacaktır.

S.1.



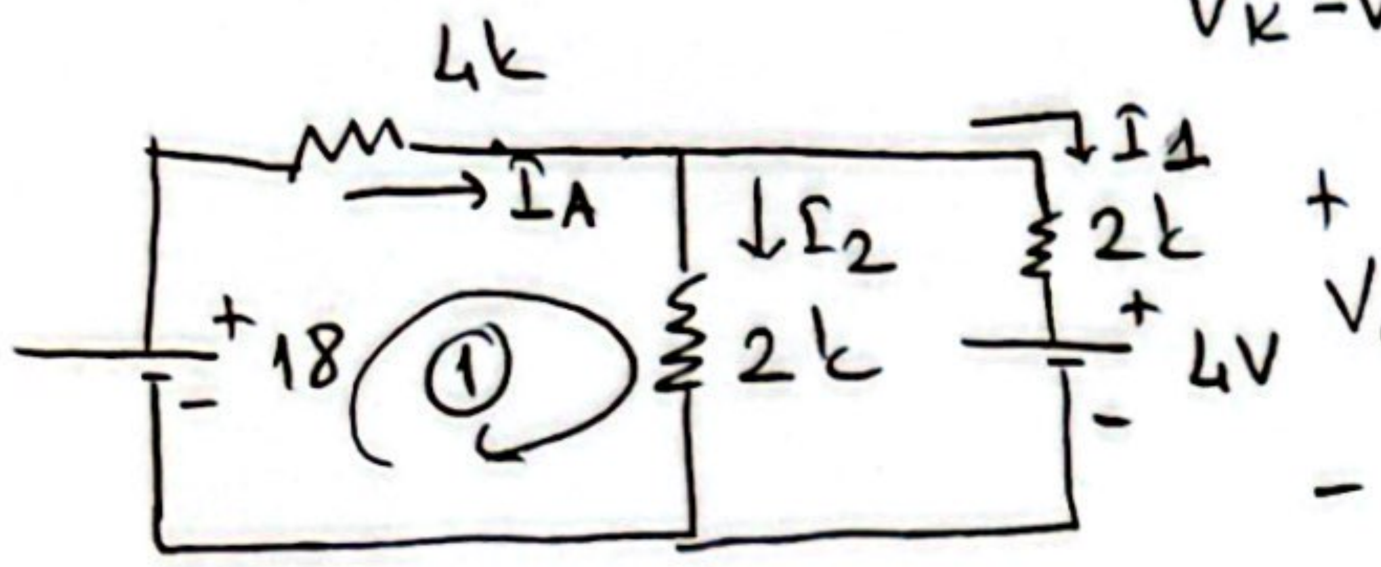
Şekildeki devrede V_i=18V Vz=4V olarak verilmektedir. Buna göre V_o voltajını ve I_2 akımını bulunuz.

Çözüm:

⇒ Diyot iletimde $(V_A = 18V, V_K = 0V)$ (2p)

⇒ Zener için; $V_A = 0 \quad V_K = \frac{18}{6k} \cdot 2k = 6V$

$V_K - V_A \geq 4V$ old. için ters yönde iletimde (3p)



$I_A = I_2 + I_1$

$2k \cdot I_2 = 2k \cdot I_1 + 4$

$2k(I_2 - I_1) = 4$

$I_2 - I_1 = 2mA \Rightarrow \boxed{I_1 = I_2 - 2mA}$

$I_A = I_2 + (I_2 - 2mA) \Rightarrow \boxed{I_A = 2I_2 - 2mA}$

1 nolu çevre:

$18 = 4k \cdot I_A + 2k \cdot I_2$

$18 = 4k(2I_2 - 2mA) + 2kI_2 \Rightarrow 18 = 8kI_2 - 8 + 2kI_2$

$26 = 10kI_2$

$\boxed{I_2 = 2{,}6mA}$ (10p) $\rightarrow I_1 = I_2 - 2mA$

$I_1 = 0{,}6mA$

$V_o = 2k \cdot I_1 + 4$

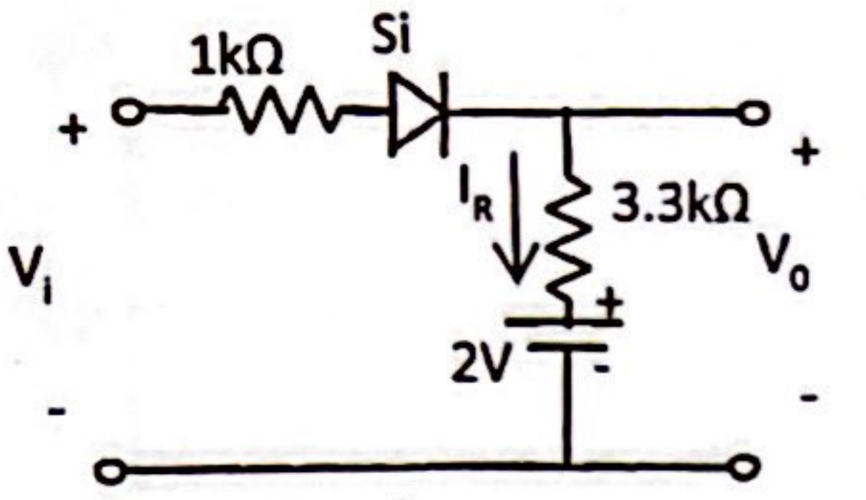
$V_o = 2k \cdot 0{,}6mA + 4$

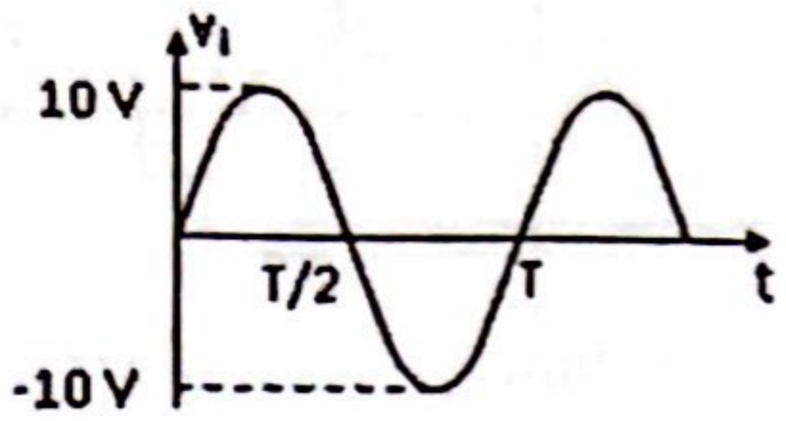
$\boxed{V_o = \cancel{?{,}?}\ V}$ 5,2 V

(10p)

Scanned with CamScanner

S.4.





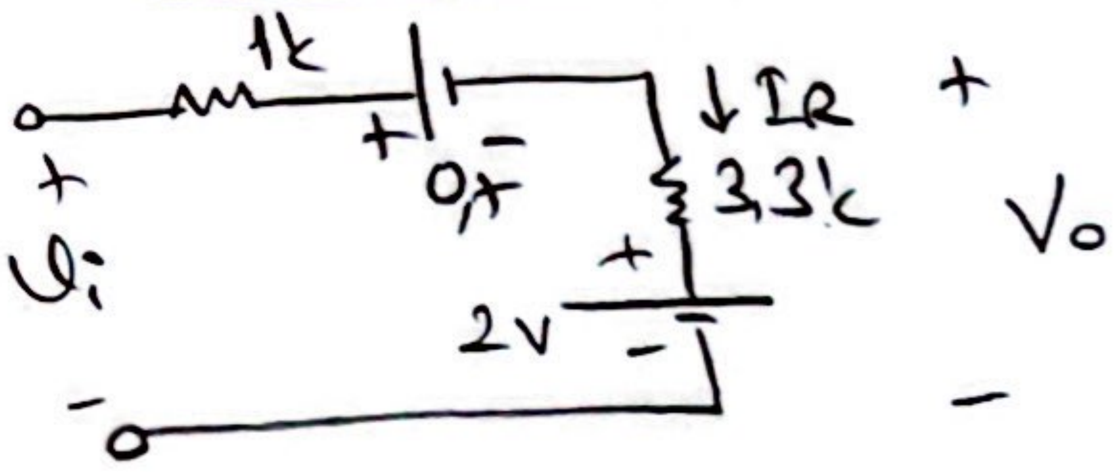
Yandaki devre için V_0 ve I_R nin dalga şekillerini ölçekli olarak çiziniz.

Si Diyod için:

$V_A = V_i$, $V_K = 2V$ $\Rightarrow$ $V_i - 2 \geq 0,7$ ise D iletimde

$\boxed{V_i \geq 2,7 \text{ ise Diyod iletim} \quad < 2,7 \text{ ise Diyod kesimde}}$

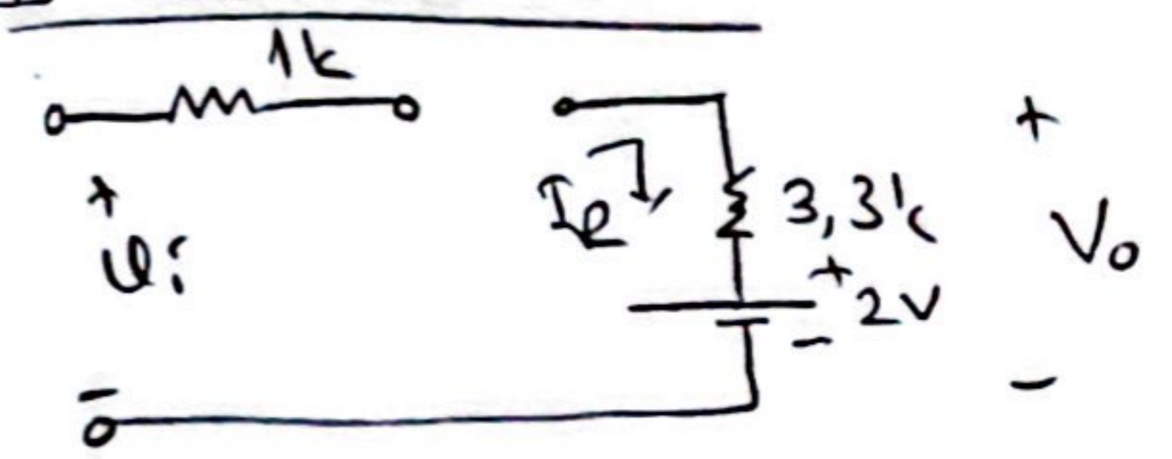
D iletimde iken:



$\boxed{I_R = \frac{V_i - 2,7}{4,3k}}$

$V_o = I_R \cdot 3,3k + 2 \Rightarrow \boxed{V_o \cong 0,76 V_i}$

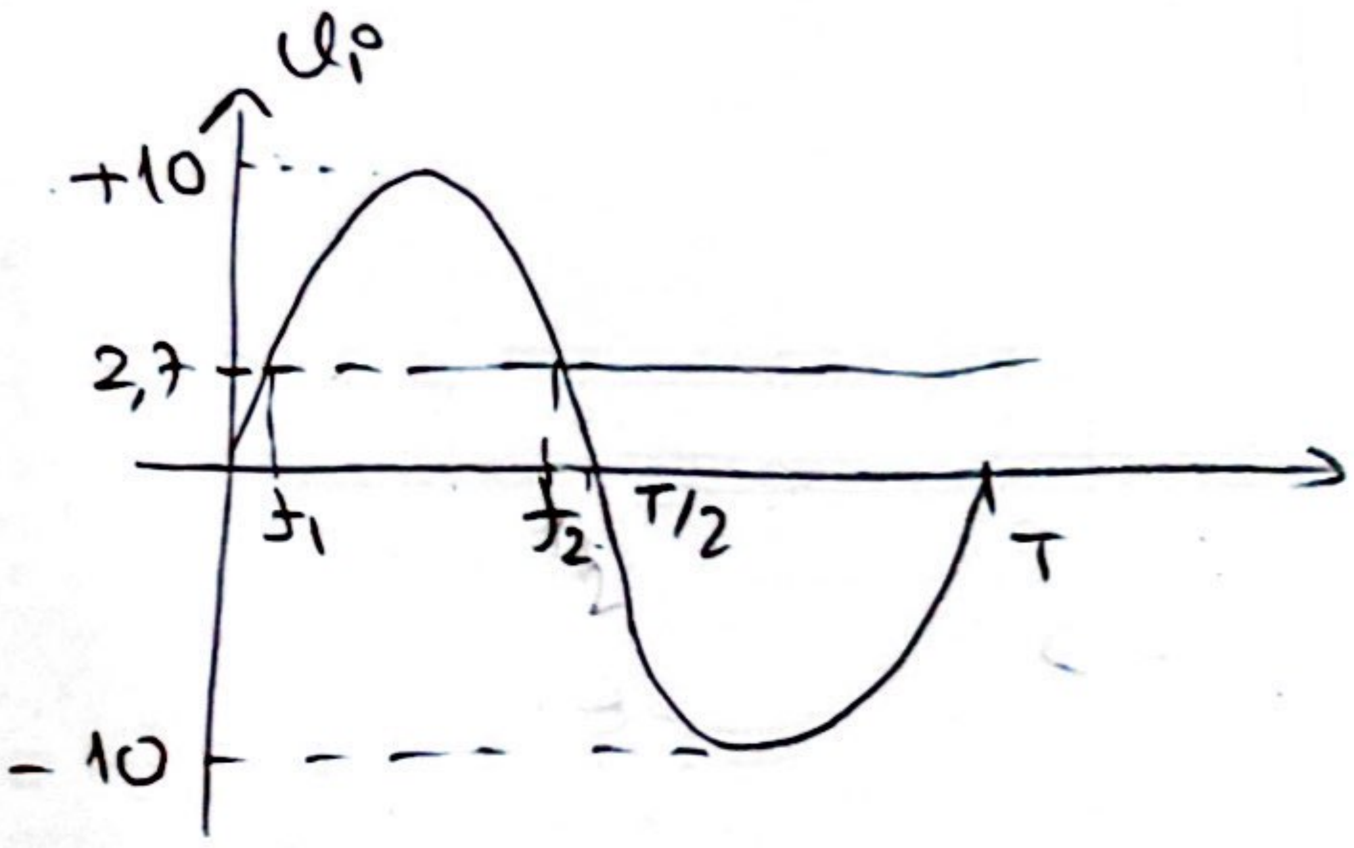
D kesimde iken:



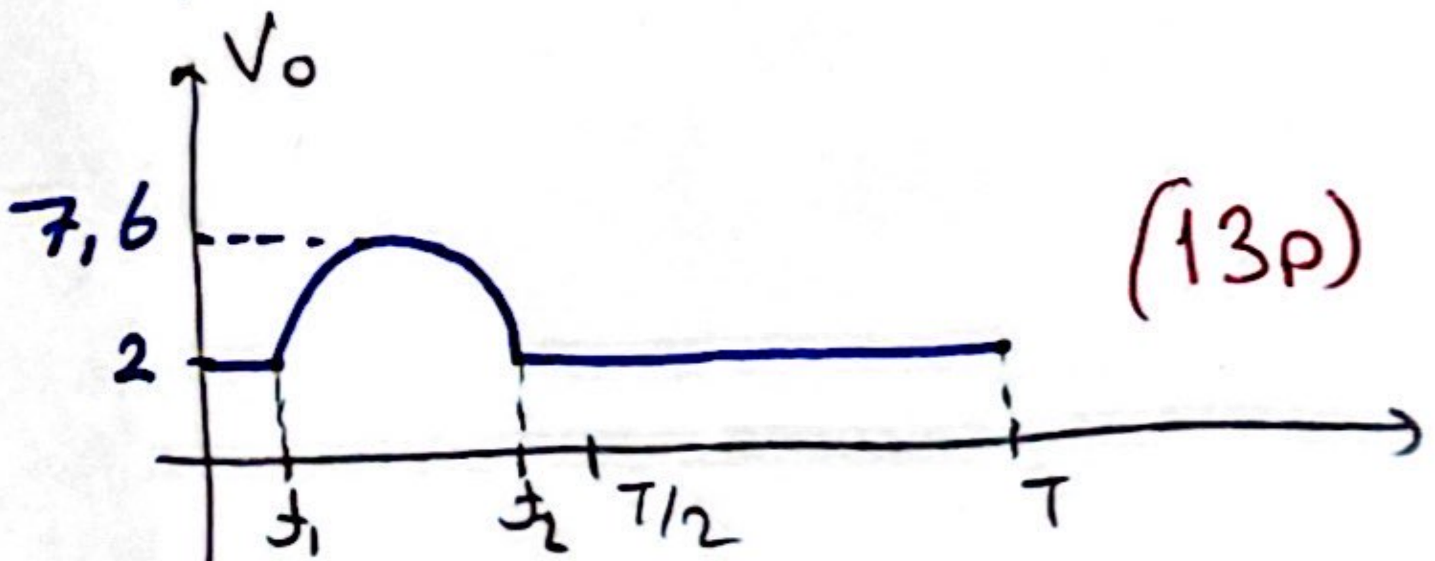
$\boxed{I_R = 0}$

$\boxed{V_o = 2}$

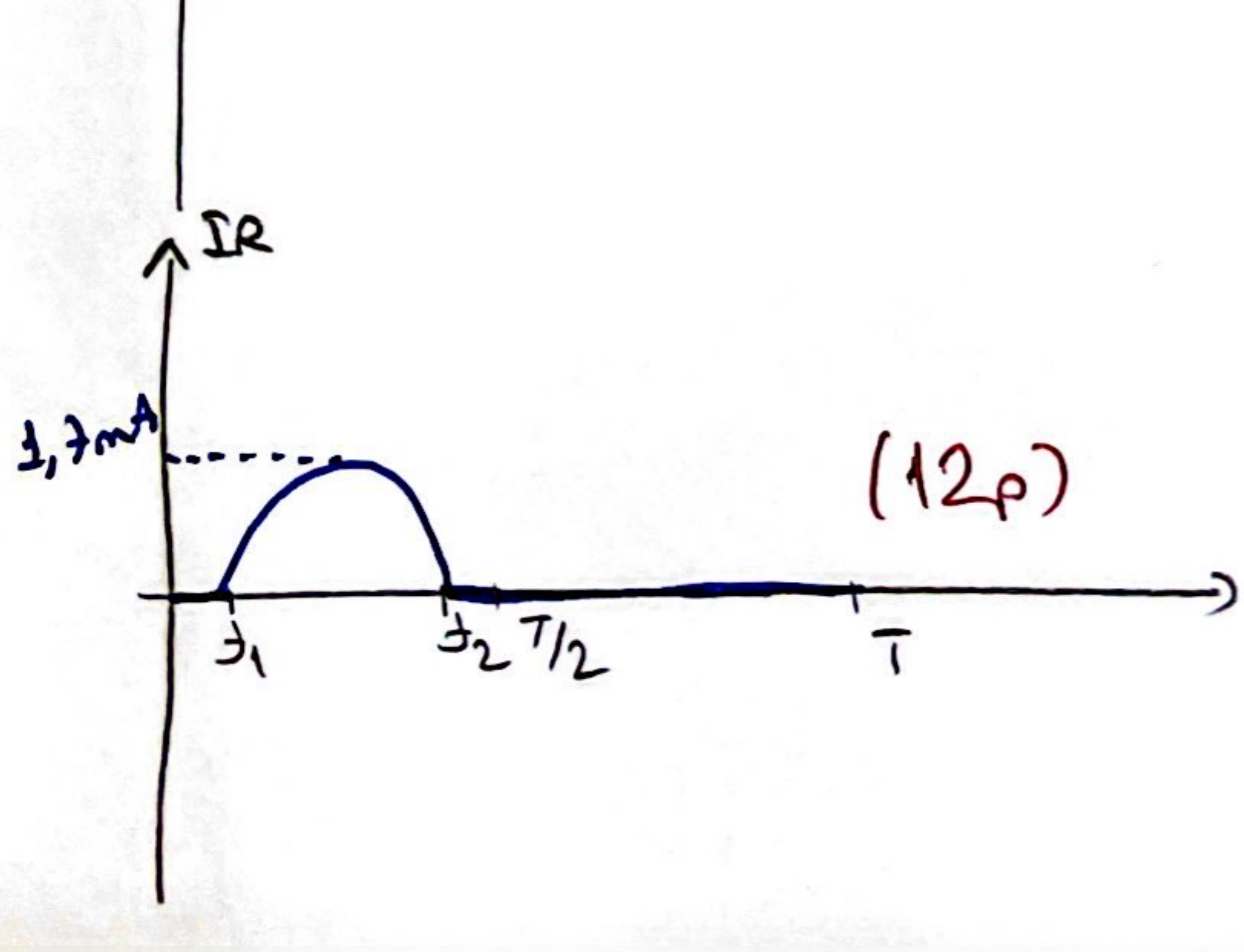
girdiş yolu.



Not:
* girdiş yolu doğru dalga şekillerinin çizimi yoksa) ⇒ (15p.)

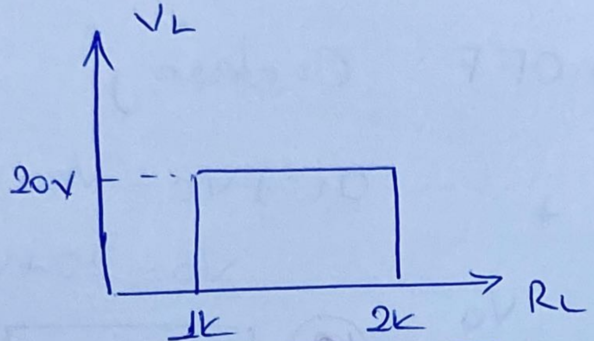


(13p)



(12p)

(2)



Soruda Verilenler ⇒

$V_Z = 20V \quad R_{L_{min}} = 1k$

$R_{L_{max}} = 2k$

$V_i = 60V$

$$R_{L_{min}} = R \cdot \frac{V_Z}{V_i - V_Z}$$

$$1k = R \cdot \frac{(20)V}{(60-20)V}$$

$$\boxed{R = 2k}$$ 5 puan

$V_R = V_i - V_Z$

$V_R = 60 - 20 = 40V$

$$I_R = \frac{V_R}{R} = \frac{40V}{2k} = 20mA \Rightarrow \boxed{I_R = 20mA}$$ 5

$$I_R = I_{Z_{max}} + I_{L_{min}} \Rightarrow I_{Z_m} = I_R - I_{L_{min}}$$

$$I_{Z_m} = 20mA - 10mA$$

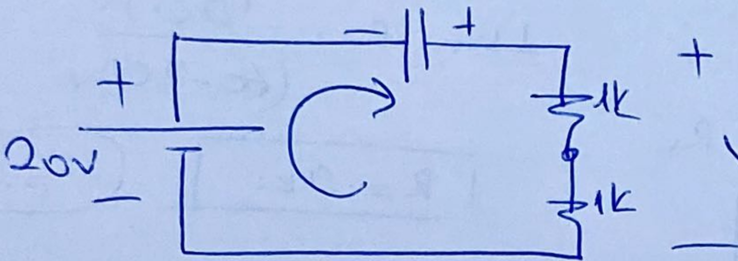
$$I_{L_{min}} = \frac{V_Z}{R_{L_{max}}} = \frac{20V}{2k} = 10mA \Rightarrow \boxed{I_{Z_m} = 10mA}$$ 5

$$\boxed{I_{L_{min}} = 10mA}$$ 5

$$P_{Z_m} = V_Z \cdot I_{Z_m} = 20V.(10mA) = 200mW$$

$$\boxed{P_{Z_{max}} = 200mW}$$ 5

3 (+) alternansta; D:OFF C: deşarj

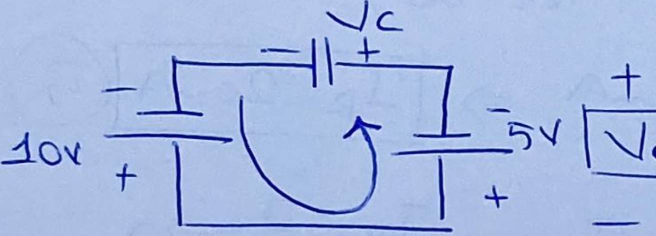


$20 + V_C - V_o = 0$

$V_o = 20 + V_C$

10 $\boxed{V_o = 25V}$

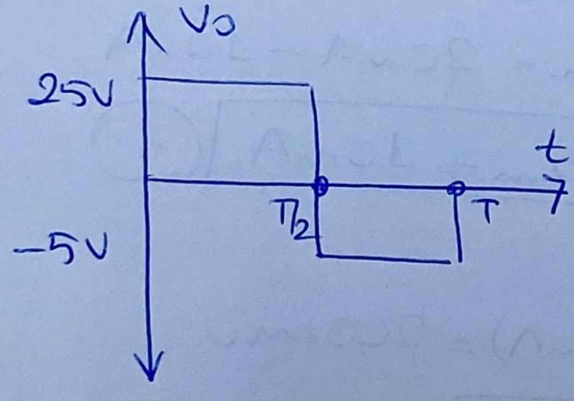
(−) alternansta; D:ON C: şarj



10 $\boxed{V_o = -5V}$

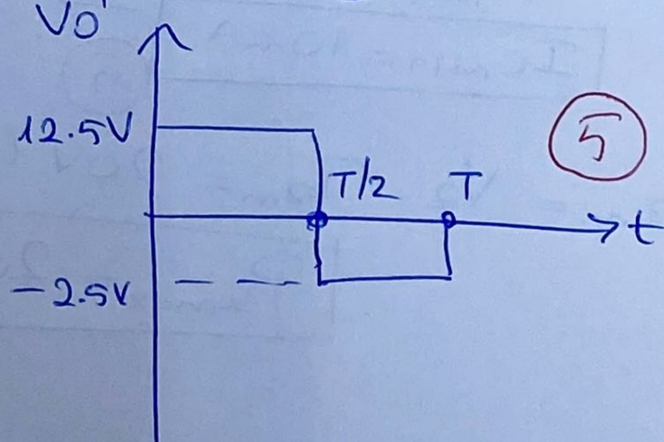
$10 - 5 - V_c = 0$

$\boxed{V_c = 5V}$



Soruda istenen

$V_o' = \dfrac{V_o}{2}$ old. göre;



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